

AdaBoost algorithm

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1 The algorithm

0. We have N data points, assign each data point the same weight $\alpha_i = 1/N$

We have T weak learners (decision stumps)

For $t = 0, 1, \dots, T-1$

1. Learn weak learner f_t
2. Compute weighted error ε_t
3. Compute weight w_t
4. Recompute α_i for $i = 0 \dots N - 1$
- 4a. Normalize α so that

$$\sum_{i=0}^{N-1} \alpha_i = 1$$

2 Formula for 2. weighted error

$$\varepsilon_t = \sum_{i=0}^{N-1} \alpha_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$$

3 Formula for 3. weight

$$w_t = \frac{1}{2} \ln \frac{1 - \varepsilon_t(f_t)}{\varepsilon_t(f_t)}$$

Intuition: w_t measures the importance of the current weak learner. A lower weighted error gives it a larger weight.

4 Formula for 4. new α

$$\alpha_i^{\text{new}} = \begin{cases} \alpha_i \cdot e^{-w_t} & \text{if } \hat{y}_i = y_i \\ \alpha_i \cdot e^{w_t} & \text{if } \hat{y}_i \neq y_i \end{cases}$$

Intuition: This is increasing the weight for data points that have been misclassified by current learner.

5 Prediction function

For a given feature matrix X_{test} and data point i

For each weak learner

predict using the learner

multiply outcome by the weight of the learner

Sum that over all learners

Get the sign of the outcome: + then +1 - then -1

6 Prediction formula

$$\text{total}_i = \sum_{t=0}^{T-1} w_t \cdot f_t(X_{\text{test},i})$$
$$\text{outcome}_i = \begin{cases} -1 & \text{if } \text{total}_i < 0 \\ +1 & \text{if } \text{total}_i \geq 0 \end{cases}$$

7 Prediction matrix-wise

This can be implemented using matrix operations:

- Predict all data points with each weak learner, producing an $N \times T$ matrix Y_{combined} .
- Store all learner weights in a $T \times 1$ matrix $\mathbf{W}$.
- Compute the totals:

$$\mathbf{total} = Y_{\text{combined}} \mathbf{W}$$

where $\mathbf{total}$ is an $N \times 1$ matrix.

- Convert each total to -1 or $+1$:

$$\mathbf{outcome}_i = \begin{cases} -1 & \text{if } \mathbf{total}_i < 0, \\ +1 & \text{if } \mathbf{total}_i \geq 0. \end{cases}$$